

ETL Architecture Design and Star Schema Modeling for the Deep Work Disruption Dataset

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Abstract—This project designs and implements an automated ETL pipeline for the `social_media_usage_mental_health.csv` dataset using Pentaho Data Integration (PDI). The pipeline runs a full load ingestion strategy and validates incoming data through three layers: missing value removal, row deduplication, and string standardization. The transformation layer also builds new categorical variables for age demographics and fatigue levels. Pentaho Jobs orchestrate the extraction and load sequence, and the pipeline restructures the flat file into a Star Schema with three dimension tables and one fact table, producing a dataset ready for predictive modeling.

Keywords— Deep Work, Machine Learning, Data Engineering, Short-form Content, Generation Z

I. INTRODUCTION

Predictive modeling depends on high-integrity data to maintain accuracy and prevent model drift. Analytics projects that study deep work disruption and mental fatigue draw on raw data from public sources, and that data rarely carries any guarantee of structure. Researchers often use pre-cleaned datasets for isolated machine learning tasks, but a data engineering approach built for production needs infrastructure that processes raw data on its own. An automated ETL pipeline built with Pentaho Data Integration (PDI) ingests, cleans, and standardizes flat files before analysis or model training begins.

This project uses the public `social_media_usage_mental_health.csv` dataset to address the structural and semantic vulnerabilities of unvalidated flat files. A public dataset might look structured on download, but a data engineering pipeline has to treat data purity as unproven: it has to catch undocumented missing values, duplicate entries, and inconsistent string formatting such as stray spaces or mixed casing that corrupt categorical fields. Flat-file structures also perform poorly for multi-dimensional queries and business intelligence dashboards. Without an ETL architecture that transforms these raw files into a relational schema, downstream operations face data leakage, inconsistent feature engineering, and slow queries.

This project builds an automated, full load data ingestion and transformation pipeline in PDI. The first objective builds data quality layers that filter nulls, standardize categorical strings into lowercase, and deduplicate sorted records. The second objective covers feature engineering: new categorical ranges for age demographics and mental fatigue disruption levels. The pipeline then restructures the flat CSV data into a Star Schema with four dimension tables and one fact table, with Pentaho Jobs orchestrating the full workflow for automated execution.

II. LITERATURE REVIEW

Accurate predictive modeling requires high-quality data, since raw data assets carry inconsistencies and structural vulnerabilities by default. Yin (2025) argues that automated ETL processes give data quality control its main mechanism, removing missing values, noise, and semantic anomalies before analytical models run [1]. Kumar et al. (2023) add that raw data collection often suffers from operational errors and poor documentation, and they show that validation rules built directly into the ETL pipeline improve data reliability, prevent data leakage, and support sound decisions downstream [2]. An automated pipeline that ingests and cleans flat files works as a preventative layer, not an administrative step.

Structuring processed data for analytical efficiency matters as much as ingestion and validation. Tan et al. (2024) test a centralized Data Warehouse built on a Star Schema and find it effective in practice. Splitting flat structures into a central fact table and several descriptive dimension tables organizes the relational environment and speeds up complex queries and business intelligence operations [3]. Converting a flat, unoptimized file into a Star Schema keeps categorical integrity intact across user demographics and time attributes, which optimizes the data layer for the machine learning tasks that follow.

Applying machine learning to social media logs, in the domain of digital behavior and mental workload, demands a carefully engineered feature space. Selvakumar et al. (2025) build a machine learning approach to predict social media usage behavior and find strong predictive power in demographic

attributes like age range alongside engagement metrics like daily time spent [4]. Their findings support this project's approach: standardizing behavioral features, including content types, platform categories, and mental fatigue thresholds, within the ETL transformation layer produces reliable, empirical insight into generational digital disruption.

III. SYSTEM DESIGN

A. System Architecture

In this stage, raw data from a comma-separated-values file loaded into the staging area using MySQL Database which serves as an Online Transaction Processing (OLTP). In the transform stage, the data is processed using Pentaho Data Integration (PDI). This stage included cleaning, filtering, and relational mapping. In the load stage, the transformed data is structured into a Star Schema architecture where differentiating dimension tables and central fact table then stored into Online Analytical Processing (OLAP) as a Data Warehouse.

B. Data Ingestion

The data source for this research is a comma-separated values (CSV) file named "social_media_usage_mental_health.csv" with 5,000 rows. The raw CSV data is extracted to the system which is PDI with Full Load implementation, where the data is extracted simultaneously from the data source within each execution.

C. Data Transformation & Validation

The ETL is designed using multiple Pentaho Transformations files. One file for main transformation as in Fig 1 and several files for dimension transformations as in Fig 2.

1) Main Transformation:

- Filter Rows: To remove rows with null values in the necessary column for Machine Learning which is age and daily usage minute.
- Number Range: To make new columns or feature engineering. We make 2 new column which is fatigue category classified from mental fatigue level (eg. "Low Disruption" for "0-3", "Moderate Disruption" for "4-6", "High Disruption" for "7-10") and age category classified from user's age (eg. "Gen Z" for "16-29", "Millennials" for "30-45", "Gen X" for "46 and above")
- String Operations: To standardize text data, which includes converting strings to lowercase, removing ghost spaces, and removing digits

2) Dimension Transformation:

After the data is cleaned in the main transformation, the data is routed to separate dimension files. In these files we applied Sort and Unique Rows to remove duplicated primary keys. This ensures there are no duplicated records are loaded into the table

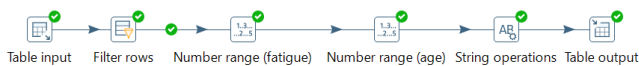


Fig 1. Main Transformation Pipeline

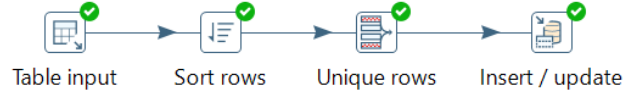


Fig 2. Dimension Transformation Pipeline

D. Dimensional Modeling

The cleaned data is transformed into a structured relational database which is Star Schema in Fig 3. This Schema is centered around one fact table named fact_usage which stores measurable metrics and foreign keys as connectors to dimension tables as shown in Fig 4. There are four dimension tables used for data filtering, grouping, and informative labeling. These tables represent users which contain user information, platform which stores social media platform information, content which stores content type information, and time to store detailed date for analyzing data and tracking trends. These dimension tables used Slowly Changing Dimension (SCD) Type 1, which means old data is overwritten with newer data, so no historical history is saved.

E. Automation & Orchestration

To manage the continuous flow of social media data, the entire process is controlled by a master Pentaho Job (.kjb) as shown in Fig 5. The pipeline is set up to run automatically on a schedule, such as every day at midnight.

This workflow runs steps in a specific order: first extracting and cleaning the data, then loading the dimension tables at the same time, and finally filling the main fact_usage table to ensure the dimension tables is properly loaded before updating fact tabel to prevent foreign keys constraint error and maintain data consistency..

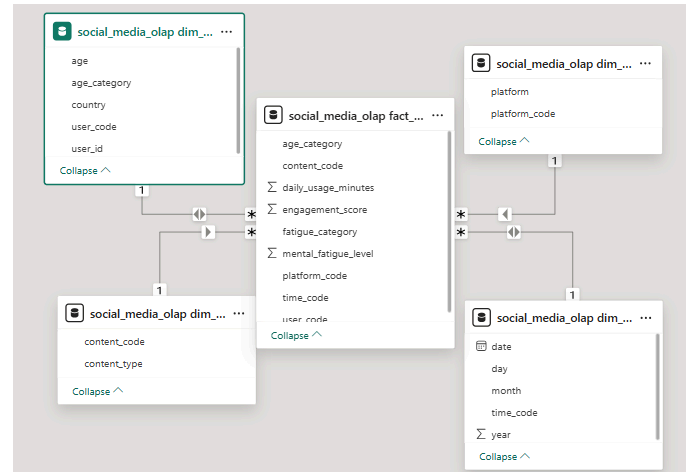


Fig 3. Star Schema Design



Fig 4. Fact Table Pipeline

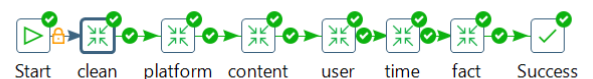


Fig 5. Orchestration Workflow

IV. RESULTS AND DISCUSSION

A. Data Consumption and Machine Learning Readiness

Following the succesful executed data pipeline, the processed data in Star Schema is extracted and served for Machine Learning analytics purpose with pipeline in Fig 6. The finalized dataset includes validated metrics and engineered variables as shown in Fig 7. As detailed in Table I and Fig. 8, the dataset provides a balanced starting point for analysis. The inclusion of control groups (Millennials and Generation X) helps prevent bias in the data.

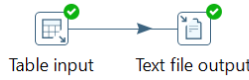


Fig 6. Export Data to Machine Learning Readiness Pipeline

#	age	age_category	country	platform	content_type	daily_usage_minutes	engagement_score	mental_fatigue_level	fatigue_category
1	54	gen x	uk	instagram	reels	253	1.0	9	high disruption
2	44	millennials	germany	instagram	reels	205	7.0	7	high disruption
3	30	millennials	uk	x	live	63	3.0	6	moderate disruption
4	58	gen x	australia	youtube	live	144	4.0	1	low disruption
5	23	gen z	usa	youtube	shorts	123	8.0	9	high disruption
6	36	millennials	pakistan	youtube	live	45	2.0	10	high disruption

Fig 7. Data Preview for Machine Learning Analytics

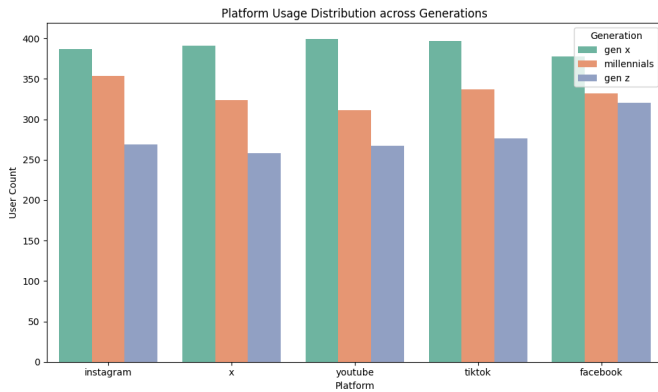


Fig 8. Feature Importance Ranking for Deep Work Disruption.

Table 1. PLATFORM DISTRIBUTION PER GENERATIONAL COHORT

Age_Categor y	Platform					
	Faceboo k	Instagra m	Tiktok	X	Youtub e	All
Gen X	378	387	397	391	399	1952
Gen Z	320	269	276	258	267	1390
Millennials	332	354	337	324	311	1658
All	1030	1010	1010	973	977	5000

B. Predictive Modeling and Behavioral Insights

The prepared data was then used to train XGBoost and Random Forest models to predict 'High Disruption' in deep work. Table II shows the performance comparison between the two models. XGBoost was chosen as the final model because it had higher precision (42.22%). Additionally, as shown by the confusion matrix in Fig. 9, XGBoost produces fewer false alarms. This proves that the dataset is strong enough to train accurate behavioral models.

Furthermore, analyzing the model's feature importance showed that the new features created during data

processing were very useful. As shown in Fig. 10, the model found that specific content types (content_type_stories and content_type_reels) combined with high screen time are the main causes of mental fatigue. The analysis also confirmed that Generation Z is more negatively affected by short-form content than older generations.

Table 2. EVALUATION MODEL RESULT

No.	Evaluation Model Result		
	Metric	Random Forest	XGBoost
1.	Accuracy	0.5300	0.5440
2.	Precision	0.4178	0.4222
3.	Recall	0.3931	0.3268
4.	F1-Score	0.4051	0.3684

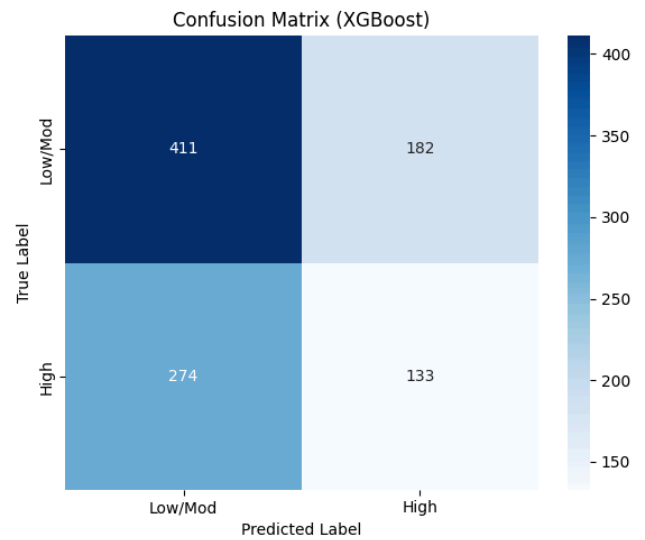


Fig 9. Confusion Matrix of XGBoost Predictions on Test Data.

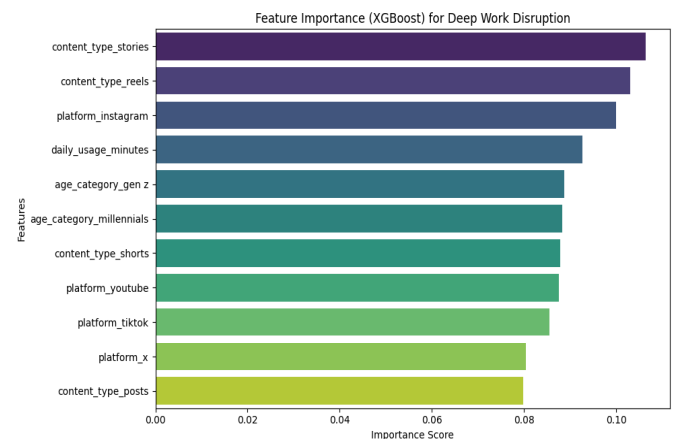


Fig 10. Feature Importance Ranking for Deep Work Disruption.

C. Technology Evaluation and Justification

This study used Pentaho Data Integration (PDI) as the main data processing engine. PDI was chosen because it facilitates rapid prototyping and modular ETL design without requiring extensive manual coding. This visual approach significantly simplifies the orchestration of complex tasks, such as managing primary key generation and referential integrity across the Star Schema.

D. Advantages, Trade-offs, and Limitations

The main advantage of this architecture is that it's neat and easy to maintain. This is achieved by clearly separating the data cleaning process from the data loading process.

By implementing Slowly Changing Dimension (SCD) Type 1 for all dimension tables was the best choice to save storage space and speed up queries. It simply overwrites old data with new data, avoiding the wasted memory and processing power of tracking historical changes.

The downside of this design is its reliance on a Full Load method (pulling all data from scratch). Extracting all 5,000 rows at once is fast and easy for now, but it will slow down the system if the data grows significantly in the future. For future improvements, an incremental load method (only loading new or updated data) can be used to handle large amounts of data more efficiently.

V. CONCLUSION

In conclusion, this project successfully built an automated data pipeline to analyze how social media affects mental health. Using Pentaho Data Integration (PDI), the data was extracted from a CSV file in a single step. The data was then cleaned, and new useful categories were created, such as the user's generation and mental fatigue level.

Next, the data was organized into a Star Schema. This structure includes one main table (fact_usage) and three supporting tables (dim_user, dim_time, and dim_platform_content). The SCD Type 1 method was applied to these dimension tables to save storage space. To ensure the steps ran smoothly in the correct order, Pentaho Jobs was used to control the entire process.

Ultimately, a clean and well-structured dataset was produced. As demonstrated in the Jupyter Notebook, this final data is fully prepared and ready to be used immediately for training machine learning models.

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